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An Optimization Framework for Soft Fluidic Actuators: Fully Coupled FSI Simulation with Surrogate Modeling

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***Abstract*—Soft fluidic actuators (SFAs) are promising for underwater manipulation and embodied soft systems, but they are often modeled and designed using structural finite element analysis that neglects coupled fluid loading, which can bias the pressure–deformation response that the design process relies on. This paper introduces a two-way external fluid–structure interaction (FSI) framework for geometric optimization of underwater SFAs, combining coupled simulation with surrogate response surfaces and constrained search over a parameterized design space. We first quantify the modeling gap by comparing baseline structural FEM predictions with two-way external-FSI predictions for the same geometry. We then train surrogate nonparametric regression (NPR) response surfaces from a broad design-of-experiments dataset and perform two optimizations, one driven by FEM-only data and one driven by FSI data, followed by high-fidelity two-way coupled verification. Experimental validation shows that the resulting FSI optimum achieves similar bending performance at 30% lower pressure relative to the FEM optimum while increasing bending angle by 46%. System-level tests confirm these gains transfer to practice, with pull-force measurements showing a 58% mean improvement in retention force and a fish-like body with paired actuators achieving 24.8% higher forward velocity.**

***Index Terms*—Soft fluidic actuators, fluid–structure interaction, surrogate modeling, design optimization, underwater robotics**

I. Introduction

SOFT fluidic actuators (SFAs) enable large deformation and inherent compliance, which makes them attractive for underwater manipulation and bio-inspired systems requiring conformal contact. Recent reviews emphasize progress in soft-fluidic-actuator materials, fabrication, modeling, control, and underwater applications, while also noting that predictive design remains difficult because the response depends strongly on geometry, hyperelastic material behavior, fabrication variability, and environmental loading [1]–[3]. Despite their prevalence, most SFA modeling and design pipelines rely on finite element modeling (FEM) or simplified analytical approximations in which the surrounding fluid is neglected or treated implicitly [1], [4], [5]. While these approaches can be useful for rapid iteration, they do not explicitly capture coupled fluid loading that can alter deformation paths, energetic cost, and stress distributions during underwater operation.

Two-way fluid–structure interaction (FSI) modeling has therefore gained attention as a more physically grounded tool for underwater SFAs, because it explicitly resolves the coupled exchange between actuator deformation and fluid loading [6]. Existing FSI-related works include studies of internal chamber flow and pressurization dynamics [6], [7] and external-flow FSI where the surrounding water induces additional hydrodynamic loading [8], [9]. These studies demonstrate the importance of coupled loading for soft actuator prediction, but they primarily use FSI for modeling or characterization of prescribed geometries rather than for selecting optimized actuator dimensions.

Parallel to modeling, soft actuator design optimization has advanced through parameter optimization, topology optimization, surrogate-based design, and learning-based approaches. FEM-based studies have optimized pneumatic or fluidic actuator dimensions for bending, force, or task performance [4], [5], [10], [11], while topology optimization has expanded the design space for soft grippers and pneumatic actuators [12]–[14]. Surrogate and learning-based methods further reduce the cost of repeated high-fidelity simulations and have been used for model-based optimization, reinforcement-learning-assisted design, and shape matching [15]–[18]. While these methods can yield improved geometries, they are typically developed and validated under FEM-only assumptions and, in many cases, in air-only conditions [5], [12], [13], [15], [17]. For underwater actuation, this separation is consequential because FSI effects, including hydrodynamic drag, added mass, and viscous damping, modify the pressure–response mapping and can shift the coupled-loading optimum relative to FEM-only predictions. Consequently, designs selected by FEM-based optimization may be biased when transferred to water. Overall, the literature largely divides into FSI studies focused on modeling and characterization [6]–[9] and optimization frameworks that do not include explicit external fluid coupling during design selection [5], [10], [12], [13], [15], [17]. What remains under-examined is the value of placing two-way FSI inside the optimization loop and validating the resulting design choices experimentally in water, including robustness under

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external flow.

This paper addresses this gap with an external-FSI-in-the-loop optimization and validation framework for underwater soft fluidic actuators, where internal actuation is prescribed as pressure and the resolved coupling is between actuator deformation and the surrounding water. We first compare a baseline structural FEM model with the same structural model coupled to the surrounding water through two-way external FSI, thereby quantifying the prediction bias caused by neglecting external fluid loading. We then build surrogate response surfaces from a broad Design of Experiments (DOE) over the feasible design space and perform constrained optimization using both FEM-only and FSI-derived datasets to isolate the impact of modeling fidelity, followed by high-fidelity two-way FSI verification of the resulting optima. Experimental validation is conducted in water through bending, blocked-force, and pull-force characterization of the FEM-optimized and FSI-optimized designs, with corresponding two-way FSI predictions for comparison. Robustness is evaluated in a controlled swim tunnel under external flow, and system-level relevance is demonstrated by integrating optimized actuators into a three-finger gripper for underwater grasping and by assembling paired actuators into a simple fish-like body driven by antagonistic pumping. Together, these results provide an experimentally grounded pipeline that places FSI inside the design loop and shows that geometry selection based on coupled fluid loading yields improved underwater performance relative to FEM-only optimization. The main contributions of this work are threefold.

1) An external-FSI-in-the-loop framework that incorporates coupled responses into geometry selection, embedding hydrodynamic loading in the optimization data and making fluid-only quantities such as drag available for design evaluation.
2) A paired cross-evaluation showing that omitting external fluid loading introduces geometry-dependent bias that reorders candidate designs rather than uniformly rescaling their responses, thereby changing the selected optimum.
3) Controlled hardware evidence that FSI-informed geometry selection improves underwater performance over FEM-based selection, with robustness under external flow and gains retained after integration into gripper and fish-like assemblies.

The remainder of this paper is organized as follows. Section II introduces the actuator design and fabrication. FSI simulation framework and the optimization methodology are explained in Section III and Section IV, respectively. Section V presents the simulation and experimental results, including actuator characterization in quiescent and flowing water as well as gripper and fish-like body demonstrations. Section VI concludes the paper.

II. Design Parameters and Actuator Geometry

The fluidic actuator in this study follows the well-established PneuNet design [19]. PneuNet actuators are widely adopted in soft robotics due to their rapid response, simple structure, and ease of fabrication [4], [20], [21]. Their extensive prior use and available experimental benchmarks make them a suitable platform for evaluating the proposed methodology and for direct comparison with existing studies.

As shown in Fig. 1(a), the actuator comprises a chambered main body bonded to a two-layer base with an embedded inextensible layer. The main body contains 11 corrugated chambers, each $10\times8\times15$ mm (axial length × width × height), giving a total actuator length of 110 mm. The chamber side walls have thicknesses $(t_{w1} = t_{w2} = 1)$ mm, while the upper wall is 2 mm to promote bending without twisting. To account for higher pressures near the inlet, the inlet wall thickness is increased to 2 mm while the remaining side walls remain 1 mm [6]. The chambers are supplied through a central duct of width $w_c = 2$ mm and height 2 mm. The top and bottom base layers are each 2 mm thick, with a 0.1 mm inextensible paper layer embedded between them to suppress axial extension. The design variables considered in this study are the chamber sidewall thicknesses (t_{w1}, t_{w2}) and the channel width w_c. The actuator with the aforementioned dimensions serves as the baseline for further analysis. The complete multi-step molding process for actuator fabrication is shown in Fig. 1(b).

III. FSI Simulation Setup

Accurate closed-form modeling of SFAs operating underwater is difficult due to large-strain hyperelasticity, geometric nonlinearity, and fluid loads that change with the evolving actuator shape. To capture these coupled effects, we employ two-way FSI simulations to generate pressure–response input–output maps and the associated performance metrics that guide geometric refinement and surrogate-based parameter optimization. FSI modeling captures the coupled interaction between a viscous flow and a deformable solid across a shared interface [22]. Fluid forces induce stresses and deformation in the structure, and the resulting deformation modifies the flow field, so both the structural and fluid unknowns must be resolved. When deformation-induced changes to the flow are negligible, a one-way coupling is sufficient, in which fluid loads are applied to the structure without updating the fluid solution. For the large deformations typical of SFAs, the feedback to the fluid is non-negligible and a two-way, fully coupled formulation is required [6]. In our implementation, the fluid is solved in Fluent using the three-dimensional, viscous, incompressible Navier–Stokes equations [23]. At the fluid–structure interface, the dynamic equilibrium conditions are enforced [24]:

$$-P\mathbf{n} + \boldsymbol{\tau}_{\text{eff}} \cdot \mathbf{n} = \boldsymbol{\sigma} \cdot \mathbf{n}, \tag{1}$$

where P is the static fluid pressure, $\boldsymbol{\tau}_{\text{eff}}$ is the effective viscous stress tensor of the fluid, $\boldsymbol{\sigma}$ is the Cauchy stress tensor of the solid, and $\mathbf{n}$ is the outward unit normal vector.

A. Hyperelastic Material Model

SFAs undergo substantial deformations during pressurization. Hence, the silicone rubbers used in their fabrication are typically modeled as hyperelastic materials [25], [26]. Although viscoelastic effects can be significant in certain elastomers, they are relatively minor in commonly used silicone

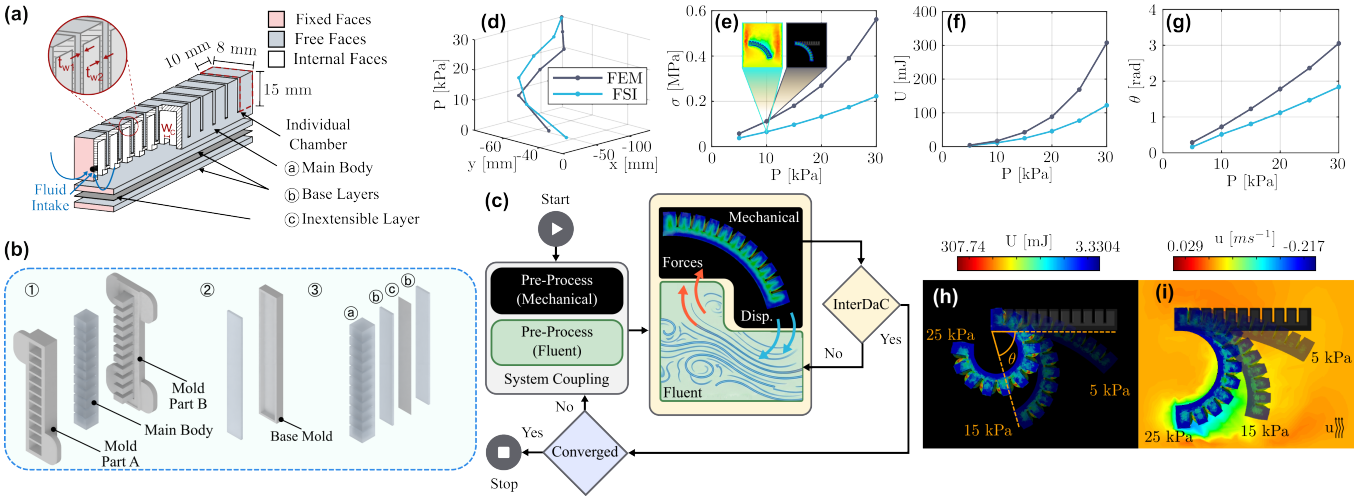

Fig. 1. **Actuator design, fabrication, FSI simulation pipeline, and baseline FEM–FSI comparison.** **(a)** PneuNet schematic showing chambers, base layers, inextensible layer, and key geometric parameters. **(b)** Molding workflow: ① chambered-body mold and casting, ② base-plate mold and casting, and ③ final assembly with the inextensible layer. **(c)** Mechanical and Fluent are iteratively solved with exchange of interface displacements (Disp.) and fluid forces until interface-data convergence (InterDaC) and overall convergence are satisfied. **(d)** Tip trajectory of the baseline actuator over the pressure sweep. **(e–g)** Responses of the baseline actuator versus pressure under uniform upstream flow $v_s = 0.01$ m/s: (e) maximum equivalent stress $\sigma_{\max}$ with stress-field insets, (f) strain energy U, and (g) bending angle θ, showing structural FEM overpredicts deformation and associated metrics relative to the corresponding external-FSI simulations. **(h)** FEM-based deformed shapes and strain-energy distributions of the baseline actuator at representative pressures used to compute U and define θ. **(i)** Two-way FSI-based deformation shapes and strain-energy distributions of the baseline actuator with velocity field under uniform upstream flow $v_s = 0.01$ m/s (arrow indicates flow direction).

rubbers [27]. In this study, DragonSkin 20 silicone rubber (Shore hardness 20A, density 1.08 g/cm³) is used to make the SFA. Its behavior is captured using the incompressible third-order Yeoh model [26], [28], whose strain energy density is

$$W = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2 + C_{30}(\bar{I}_1 - 3)^3, \qquad (2)$$

where $\bar{I}_1 = J^{-2/3}(\lambda_1{}^2 + \lambda_2{}^2 + \lambda_3{}^2)$ is the modified first invariant of the Cauchy–Green deformation tensor, λ_i are the principal stretch ratios, and J is the volume ratio between deformed and undeformed states (equal to 1 for incompressible materials). The derivation of the stretch-invariant framework, the general N-term Yeoh formulation, and the stress–stretch relation are provided in Supplementary Section 1.1. The parameters C_{10}, C_{20}, and C_{30} are obtained via mechanical property tests (see Supplementary Section 1.1 and Fig. S1 for testing details and results). The strain-limiting layer, made of paper, is characterized by Young's modulus $E = 1.2$ GPa, Poisson's ratio $\nu = 0.2$, and thickness $\delta = 0.1$ mm.

B. FSI Simulations for Actuator

The actuator response is simulated using two-way FSI with chamber pressure ramped from 0 to the design pressure $P_d = 10$ kPa at $t = 1$ s. Internal flow and pressurization dynamics are not modeled [7], [8]; the prescribed pressure is applied as a chamber-wall load, and the two-way coupling is restricted to external water loading on the actuator surface. This approximation is appropriate for the present quasi-static design comparison because prior internal-flow FSI studies show that, under slow pressurization, internal pressure gradients become small and the deformation approaches the uniform-pressure response, whereas internal-flow FSI is mainly needed for high-speed transient inflation [6], [7]. The surrounding water is modeled as incompressible flow with a uniform upstream velocity v_s. Solid and fluid domains were created in SolidWorks and imported into ANSYS Workbench 2024 R2 (ANSYS Inc., Canonsburg, PA, USA), where ANSYS Mechanical and Fluent were coupled via System Coupling.

Structural simulations were performed in ANSYS Mechanical using Transient Structural analysis. The solid domain was meshed after suppressing the fluid geometry using a tetrahedral mesh with nonlinear mechanical preference and 1 mm element size. The strain-limiting paper layer was modeled as a shell body. The actuator inlet face was fixed, corresponding to the pink-shaded face in Fig. 1(a). Submersion was modeled by assigning a fluid–solid interface on the external actuator walls.

The fluid domain was solved transiently in ANSYS Fluent. The fluid volume was obtained by suppressing the solid body and meshed using a tetrahedral mesh with CFD physics preference and 2 mm nominal element size. The flow was modeled as viscous and laminar, and water properties were set to ambient conditions. The inlet boundary was prescribed as a velocity inlet with $v_s = 10$ mm/s on the wall oriented toward the bending direction, and the opposite wall was set as a pressure outlet. Remaining boundaries were treated as no-slip walls. The actuator exterior was defined as the FSI interface for coupling. FSI coupling was enforced through a dynamic mesh update of the fluid domain. Dynamic mesh smoothing and remeshing were enabled, using diffusion-based smoothing and local-cell method-based remeshing. The fluid solver used a pressure-based formulation with pressure–velocity coupling, second-order implicit time integration, and hybrid initialization. Spatial gradients were computed using the least-squares cell-based method, which assumes linear variation within each cell, offering both accuracy and computational efficiency for

the chosen mesh type [8].

The interaction between ANSYS Mechanical and Fluent was implemented through System Coupling with two-way exchange on the actuator exterior. At each coupling iteration, Fluent transferred the fluid traction on the actuator surface, consisting of pressure and viscous-stress contributions integrated as interface forces, to Mechanical. Mechanical then returned the updated interface displacements and velocities to Fluent for dynamic-mesh update of the surrounding water domain, as shown in Fig. 1(c). The two solvers were advanced sequentially at each time step until convergence. The coupled analysis was executed for $t = 1$ s with a fixed step size of 0.01 s, with automatic time stepping disabled and large-deflection settings enabled. Up to five coupling iterations were allowed per time step. The step size was chosen as the smallest value that provided stable convergence over the pressure ramp without unnecessary computational cost. A detailed description of mesh and time-step sensitivity studies is provided in Supplementary Section 1.2 and Supplementary Table S1.

1) Quantifying FEM Bias Under External Fluid Loading: Before performing surrogate-based optimization, we first quantify how neglecting fluid loading biases predicted actuator response by comparing a conventional structural FEM model with the same structural model coupled to the surrounding-water domain through two-way external FSI simulation for the same baseline geometry as discussed in Section II. The FEM model treats the surrounding medium implicitly and therefore does not capture the pressure and shear loads induced by the external flow, whereas the external-FSI model resolves the coupled exchange of interface forces and displacements. Fig. 1(d) shows the tip trajectory over the pressure sweep, indicating that structural FEM and external-FSI predictions differ even for the same actuator geometry.

Fig. 1(e–g) summarizes the resulting pressure–response differences over a 5–30 kPa sweep. Across the entire range, the structural FEM model predicts systematically larger deformation than the external-FSI model. At P_d, FEM predicts 0.76 rad while external FSI predicts 0.55 rad, corresponding to a 38% overprediction. The discrepancy grows with pressure with 63% overprediction at 30 kPa. A similar divergence is observed in the energetic and stress responses. At 30 kPa, FEM predicts $U = 307.74$ mJ and $\sigma_{\max} = 0.56$ MPa, while external FSI yields $U = 122.30$ mJ and $\sigma_{\max} = 0.22$ MPa. This shows 152% and 154% overprediction by the structural FEM model for U and σ, respectively.

Fig. 1(h) visualizes FEM deformed configurations and strain-energy distributions at representative pressures. Fig. 1(i) visualizes the velocity field around the deforming actuator in FSI simulations under a uniform upstream flow of $v_s = 0.01$ m/s, where the hydrodynamic pressure and viscous shear loads absent in the FEM-only model are reflected in both the bending profile and the surrounding flow field. Taken together, the baseline comparison motivates placing external FSI inside the surrogate modeling and optimization loop, since designs selected using FEM-only surrogates may be optimized against a biased pressure–deformation relationship under underwater loading.

IV. Optimization Framework

A. Design Variables

To improve the performance of the selected SFA and to enable systematic design exploration, we define a constrained geometric parameter space for the baseline PneuNet configuration. While the present study focuses on a standard PneuNet architecture, the same workflow applies to other actuator layouts by redefining the design variables and constraints. The geometric parameters optimized in this work are indicated in Fig. 1(a). Guided by prior analyses showing that these structural parameters strongly influence bending response [7], we parameterize the actuator using the design vector:

$$\mathbf{x} = \begin{bmatrix} t_{w1} & t_{w2} & w_c \end{bmatrix}^\intercal \tag{3}$$

B. Objective Function

In this study, the design task is posed as a model-based optimization problem to identify geometric parameters that improve deformation performance while maintaining mechanical safety. From each simulation, we extract four response variables: tip displacement, bending angle, maximum von Mises stress, and strain energy. Tip displacement is defined as the displacement of the tip face in the negative y direction under pressurization. The bending angle θ was computed using the base-to-tip coordinate method, which provides a consistent, geometry-agnostic measure for optimization and post-verification [11]. Maximum von Mises stress is evaluated on the elastomeric components only, excluding the inextensible layer, since the elastomer governs failure. Strain energy is taken as the total elastic strain energy stored in the actuator during deformation and serves as a global proxy for actuation cost. These metrics are combined into a scalar objective function as:

$$\min_{\mathbf{x}} F(\mathbf{x}) = w_1 \cdot \hat{f}_1(\mathbf{x}) + w_2 \cdot \hat{f}_2(\mathbf{x}) + w_3 \cdot \hat{f}_3(\mathbf{x}) + w_4 \cdot \hat{f}_4(\mathbf{x}) \tag{4}$$

where $\hat{f}_1(\mathbf{x})$, $\hat{f}_2(\mathbf{x})$, $\hat{f}_3(\mathbf{x})$, and $\hat{f}_4(\mathbf{x})$ denote the min–max normalized tip displacement, bending angle, maximum von Mises stress, and strain energy, respectively. The normalization is applied with the appropriate direction of improvement: tip displacement and bending angle are maximized, whereas stress and strain energy are minimized. The weighting coefficients are set to $w_1 = w_2 = 0.4$ for tip displacement and bending angle and $w_3 = w_4 = 0.1$ for stress and strain energy, reflecting the primary design goal of maximizing deformation capability while enforcing stress and energetic cost as secondary penalties (see Supplementary Section 1.3, Supplementary Table S2 and Fig. S2 for objective-weight sensitivity analysis). Eq. (4) is minimized subject to the design constraints:

$$\begin{aligned} t_{\mathrm{w1}}, t_{\mathrm{w2}}, w_{\mathrm{c}} &\in [0.5, 3.0]\,\mathrm{mm}, \\ \sigma_{\mathrm{vm,max}} &\leq 3.8\,\mathrm{MPa} \end{aligned} \tag{5}$$

The objective is intentionally defined at the actuator level rather than as task-specific objective, because task objectives would require additional assumptions about object geometry,

contact, assembly layout, and actuation policy. Performance gains from the proposed framework, however, can still be evaluated at task-level, as will be shown in later sections.

C. Design Space Sampling

To enable surrogate-based optimization, we generated a dataset that samples the feasible geometry space with broad coverage and minimal redundancy. The design vector in Eq. (3) is restricted by the fabrication constraints in Eq. (5). The admissible space was populated with $N = 90$ unique DOE points generated in ANSYS DesignXplorer after merging candidate sets and removing repeats. The DOE set combines space-filling and response-surface designs to cover the feasible region while resolving local curvature: Latin hypercube sampling was used to obtain dispersed points for linear, pure-quadratic, and full-quadratic model forms, and structured response-surface designs were added through sparse-grid initialization, Box–Behnken sampling, and multiple central composite design variants. For the Central Composite Design (CCD) family, we included max–min space-filling cycles, G-optimal and VIF-optimal selections, and enhanced rotatable and face-centered CCDs. Each sampled geometry was evaluated with the two-way FSI model to extract the performance metrics used in the aggregated objective. To isolate the impact of modeling fidelity, we reused the same $N = 90$ geometries to run FEM-only simulations without the surrounding fluid domain. This produces a paired FEM dataset under identical geometric sampling, enabling a consistent comparison between FEM- and FSI-based surrogate models and the corresponding optimized geometries.

D. Response Surface-based Surrogate Modeling

The resulting FEM-only and two-way FSI datasets were used to construct surrogate response surfaces that map the design vector in Eq. (3) to the performance metrics used in Eq. (4). First the DOE data is randomly split into 80/20 train–validation splits, with the split reshuffled across trials to reduce sensitivity to any single partition. We evaluated three surrogate families: a full second-order polynomial response surface, a single-hidden-layer neural network with 10 cells, and non-parametric regression (NPR). Here, cells refer to the number of hidden units (neurons) in the ANSYS DesignXplorer implementation of a single-hidden-layer network, which controls the model capacity. Table I summarizes surrogate validation for both datasets using the normalized root-mean-square error, expressed as a percentage of the range of the corresponding output. For a set of M samples, nRMSE is computed as

$$\mathrm{nRMSE} = 100 \times \frac{\sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(r_i - \hat{r}_i\right)^2}}{r_{\max} - r_{\min}}, \qquad (6)$$

where r_i and $\hat{r}_i$ denote the simulation output and surrogate prediction, respectively, and $r_{\max} - r_{\min}$ is the range of that output over the corresponding dataset. Lower nRMSE indicates a more accurate surrogate.

For the two-way FSI dataset, the second-order polynomial response surface exhibits consistently high error, indicating that a global quadratic form is inadequate for the nonlinear geometry–response mapping over the sampled range. The neural network achieves low error for several structural outputs, but its error remains large for the drag output F_d, which is only available from the coupled FSI simulations. NPR provides consistently low error across all reported FSI outputs, including F_d, and was therefore selected for the optimization loop. Using the same DOE geometries, we repeated the surrogate fitting for the FEM-only dataset and observed the same trend: NPR yields the most uniform accuracy, whereas the polynomial and neural-network models exhibit higher errors under identical sampling.

TABLE I
SURROGATE MODEL COMPARISON USING NRMSE (IN PERCENT) ON THE SAMPLED DATA. OUTPUTS ARE STRAIN ENERGY U, MAXIMUM VON MISES STRESS σ, TIP COORDINATES X_{tip} AND Y_{tip}, AND DRAG FORCE F_d. THE BENDING ANGLE FOLLOWS FROM THE TIP COORDINATES AND IS THEREFORE NOT FITTED AS A SEPARATE OUTPUT.

Two-way FSI			
Output	2nd Order Poly.	Neural Network	Non-Para. Reg.
U	21.14	0.39	**2.72**
σ	3.64	3.71	**2.70**
X_{tip}	4.25	0.40	**3.15**
Y_{tip}	5.03	0.74	**3.08**
F_d	40.31	36.72	**2.72**
FEM-only			
Output	2nd Order Poly.	Neural Network	Non-Para. Reg.
U	7.93	21.90	**2.82**
σ	9.46	51.13	**2.29**
X_{tip}	9.59	10.03	**2.80**
Y_{tip}	16.34	14.89	**2.93**

A more in-depth analysis for the chosen NPR response surface is reported in Fig. 2(a)–(c) for both the FEM- and FSI-generated datasets using learning CoD and nRMSE in addition to the validation nRMSE already shown in Table I. The consistently high CoD and low learning and validation error indicates that the NPR surrogate captures the dominant response trends across the outputs used for evaluation and is suitable for the subsequent optimization. Representative NPR response surfaces of the aggregated objective, constructed from the FSI dataset, are shown in Fig. 2(d)–(f). Each panel plots the objective over a pair of geometric variables with the third held fixed, visualizing how chamber wall thicknesses and channel width reshape the objective landscape under coupled fluid loading. The smooth but nonlinear structure of the surfaces further supports the need for flexible surrogate models over simple low-order approximations.

These FSI-based response surfaces define the surrogate model used within the optimization loop. Candidate designs are evaluated through the surrogate rather than by repeatedly invoking the high-fidelity two-way FSI solver, which reduces the number of expensive simulations while preserving the response trends learned from the DOE samples. In addition to the response surfaces, Fig. 2(g) reports local sensitivities of the objective and constituent outputs with respect to each design variable. This plot links geometric changes to the dominant performance drivers near the optimum, indicating which

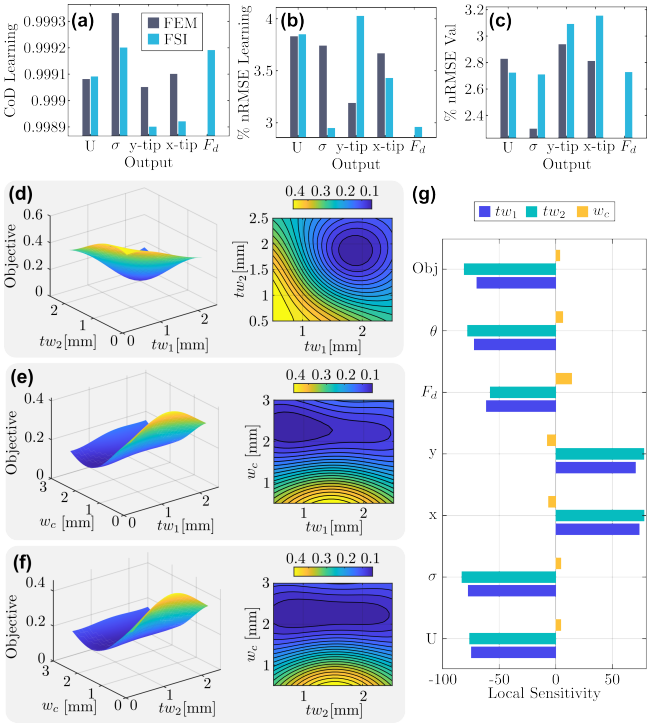

Fig. 2. **Surrogate-fit quality, FSI-based response surfaces, and local sensitivity. (a–c)** NPR surrogate fit metrics for FEM and FSI datasets: (a) learning CoD, (b) learning nRMSE, and (c) validation nRMSE. **(d–f)** FSI-based NPR objective response surfaces and contours for (d) (tw_1, tw_2), (e) (tw_1, w_c), and (f) (tw_2, w_c) with the third variable fixed. **(g)** Local sensitivity of the objective and outputs to each design variable; FEM-only surfaces are omitted for brevity.

parameters most strongly influence the output parameters. For brevity, the corresponding FEM-only response surfaces are not shown; they were generated from the same DOE geometries and are used only for surrogate and optimization comparisons.

E. Optimization Formulation and Algorithms

With the surrogate response surfaces in place, we solve for an optimal actuator geometry by solving Eq. (4) over the constraints in Eq. (5). To ensure a paired comparison, the FEM-only and FSI-based optimization workflows used the same 90 DOE geometries, design-variable bounds, actuation pressure, external-flow condition, objective formulation, and optimization procedure. The only difference was whether the surrogate-training responses were generated from conventional FEM or from two-way external FSI simulations. The full implementation details of the DOE construction, surrogate fitting, optimizer settings, FEM/FSI pairing, and final high-fidelity verification are summarized in Supplementary Table S3.

The overall workflow is summarized in Fig. 3(a). The optimization loop has four steps: first, 90 admissible geometries are generated by DOE; second, each geometry is evaluated with either the FEM-only model or the two-way external-FSI model to obtain the response metrics; third, NPR surrogate models are trained from the resulting input–output data; and fourth, the optimizer searches the constrained design space by querying the surrogate objective until convergence. Thus, the high-fidelity two-way FSI solver is used for DOE data generation and final verification, whereas the iterative optimizer itself evaluates candidate designs through the surrogate response surfaces. The optimization was performed at $P_d = 10$ kPa because this pressure represents the design actuation condition used for geometry selection (see Supplementary Table S4 for a detailed description of flow and pressure conditions).

We evaluated multiple derivative-free optimizers available in ANSYS DesignXplorer including Screening, Multi-objective Genetic Algorithm (MOGA), Non-Linear Programming by Quadratic Lagrangian (NLPQL), and Mixed-Integer Sequential Quadratic Programming (MISQP). These methods are chosen because the surrogate landscape is nonlinear and can be nonconvex, and reliable gradients are not guaranteed when the underlying high-fidelity model is multiphysics and strongly nonlinear [29]. To enable a fair comparison, all optimizers were applied to the same surrogate model over the same constrained design space and were terminated using a common relative convergence threshold of 2% on the objective. The Screening method performs a coarse global search to identify promising regions followed by refinement; it was configured for 3000 surrogate evaluations with a maximum of 3000 generations and typically converged within about 100 generations. For MOGA, we used a population of 100 individuals per generation with a maximum of 1000 generations, and convergence was typically reached after approximately 860 generations under the same 2% criterion. NLPQL and MISQP were executed with the same convergence threshold and default DesignXplorer settings for step control and con-

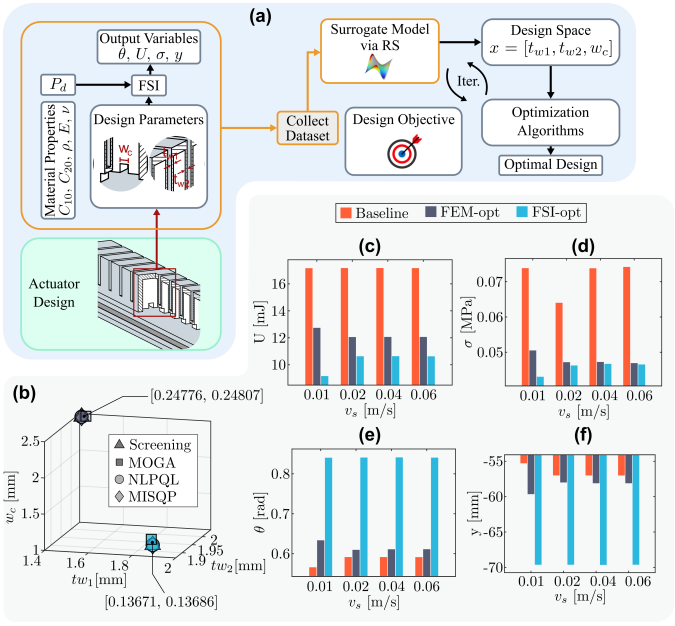

Fig. 3. **Optimization workflow, optimizer consistency, and two-way FSI verification under external flow at P_d. (a)** Surrogate-based optimization loop from DOE sampling and response-surface training to iterative search. **(b)** Optima from multiple optimizers in (tw_1, tw_2, w_c); brackets show the objective range across optimizers for FEM- and FSI-based cases. **(c–f)** Two-way FSI verification of baseline, FEM-opt, and FSI-opt designs over $v_s = 0.01$–0.06 m/s: (c) strain energy, (d) stress, (e) bending angle, and (f) tip displacement.

TABLE II
FINAL GEOMETRIES AND MATCHED FSI VERIFICATION RESPONSES AT 10 KPA AND $v_s = 0.01$ M/S.

Design	t_{w1} mm	t_{w2} mm	w_c mm	U mJ	$\sigma_{\max}$ MPa	Y_{tip} mm	Drag N	θ rad
Baseline	1.0	1.0	2.0	17.134	0.0737	-55.231	9.7×10^{-6}	0.565
FEM-opt	1.9	1.9	1.2	12.710	0.0501	-59.595	7.7×10^{-6}	0.633
FSI-opt	2.0	1.4	2.5	9.131	0.0429	-69.570	7.1×10^{-6}	0.840
FSI-opt vs FEM-opt	–	–	–	-28.1%	-14.4%	+16.7%	-7.80%	+32.7%
FSI-opt vs Baseline	–	–	–	-46.7%	-41.8%	+26.0%	-26.8%	+48.7%

straint handling to provide additional local-search baselines. The resulting optima obtained by the different optimizers are reported in Fig. 3(b). All optimizers converge to closely clustered designs in the space, and the bracketed annotations indicate the spread of objective values across optimizers. This consistency suggests that the surrogate-based problem admits a well-defined optimum within the admissible region and that the solution is not an artifact of a single optimizer.

V. RESULTS AND DISCUSSION

A. Simulation Results

We report two-way FSI verification results at P_d for three geometries: the baseline design, a FEM-optimized design obtained from FEM-only surrogates, and an FSI-optimized design obtained from FSI-based surrogates. The simulations were performed under a uniform upstream flow, and robustness was evaluated over $v_s = 0.01$–0.06 m/s. Table II makes the FEM-opt/FSI-opt comparison explicit. Under the same DOE, surrogate-based optimization procedure, objective formulation, and high-fidelity FSI verification condition, the FSI-derived optimum selected a different internal geometry from the FEM-derived optimum. Compared with the FEM-opt design, the FSI-opt design increased bending angle and tip displacement while reducing strain energy, maximum stress, and the coupled drag force computed on the pressurized, deformed actuator. The drag difference here is not attributed to a different undeformed external footprint; rather, the internal chamber geometry changes actuator compliance and the pressure-induced deformation state that is exposed to the surrounding flow. This indicates that including external fluid loading in the optimization loop changes both the selected geometry and the resulting underwater performance, rather than merely re-evaluating a FEM-derived optimum in water. Supplementary Section 1.4 and Table S5 isolate the origin of this shift by scoring the same sampled geometries under both datasets. The candidate preferences differ under FEM-only and coupled evaluation; across the DragonSkin 20 candidates, approximately half of all pairwise orderings change, indicating essentially no monotonic agreement between the rankings.

Figures 3(c)–(f) summarize the verified performance metrics; the design-condition values and percentage differences are listed in Table II. The FSI-optimized geometry achieves the lowest strain energy and stress while producing the largest bending angle and tip displacement among the three designs. This ordering is preserved across all tested flow speeds ($v_s = 0.01$–0.06 m/s): strain energy remains near 9.13 mJ for the FSI-optimized design versus ≈ 17.14 mJ for the baseline (46.7% reduction), and the bending angle advantage ($\approx 48\%$ over baseline) is similarly stable. At the design pressure, all stresses remain below the tensile strength of DragonSkin 20, confirming safe operation. As an additional material-sensitivity check, the same FEM-opt/FSI-opt workflow was repeated using Elastosil M4601 material parameters; the FSI-derived optimum again selected a distinct geometry and improved bending angle, tip displacement, strain energy, stress, and coupled drag force relative to the FEM-derived optimum, as summarized in Supplementary Table S6.

It is important to note here that the flow-speed trends in Fig. 3(c)–(f) are interpreted as a matched FSI verification of two selected geometries, not as a direct FEM-versus-FSI prediction-error plot. Both the FEM-optimized and FSI-optimized geometries are re-solved with the same external-FSI model at each value of v_s; therefore, increasing v_s does not selectively add fluid loading only to the FEM-optimized case. Instead, it applies the same verification physics to both designs. Over the tested range, the additional hydrodynamic perturbation does not change the geometry-controlled pressure response enough to reorder the designs or produce a monotonic growth of the FEM-opt/FSI-opt gap.

The optimization results suggest that, for underwater SFAs exposed to moderate external flow, channel width and wall-thickness asymmetry should be treated as coupled design levers rather than independent geometric details. In the present design space, external FSI shifted the selected geometry toward asymmetric chamber walls and a wider central channel; the local sensitivity analysis further indicates that w_c strongly affects the coupled drag force on the deformed actuator, while t_{w1} and t_{w2} primarily regulate bending and stress.

B. Experimental Evaluation

1) Actuator Characterization Tests: Actuator performance was characterized in a stationary glass water tank and in a recirculating Loligo swim tunnel; the testing platforms, instrumentation, and protocols are detailed in Supplementary Section 1.5 and Fig. S3(a)–(b).

The experimental sweep from 0–50 kPa includes the 10 kPa design pressure used for optimization and extends beyond it only to characterize the fabricated actuators over their usable operating range; these higher-pressure data were not used to update the optimized geometry. Because fabricated soft actuators can exhibit response variations arising from silicone batch variability, curing conditions, and wall-thickness deviations, each condition was repeated three times and is reported as mean $\pm$ one standard deviation.

Figure 4(a) compares the pressure–bending response of the FEM-optimized and FSI-optimized designs, including experimental measurements and two-way FSI predictions. Across the shared pressure range, the FSI-optimized actuator achieves larger bending at the same pressure. For example, at P_d, the measured bending angles are 0.89 rad for the FSI-optimized design and 0.61 rad for the FEM-optimized design, corresponding to a relative increase of 46%. Overall across the pressure sweep, the relative increase in bending is around 49.3%. Equivalently, to reach a target bending angle of 2.9 rad, the FSI-optimized design requires 35 kPa, whereas the

FEM-optimized design requires 50 kPa, which corresponds to a pressure reduction of 30%. The experimental trends closely follow the simulated curves with mean relative error of 3.85% when evaluated over the measured pressure points.

Blocked-force results are reported in Fig. 4(b). The FSI-optimized actuator produces higher blocked force over the tested pressure range. At P_d, the measured blocked forces are 0.1 N for the FSI-optimized design and 0.06 N for the FEM-optimized design, corresponding to an increase of 66%. Overall, mean relative difference between FSI and FEM-optimized actuator is around 43%. The FSI-optimized actuator consistently produced higher bending angles and blocked force than the FEM-optimized actuator at matched pressures, although the blocked-force variance increased at higher pressures. Representative snapshots in Supplementary Fig. S3(e)–(f) visually corroborate the measured pressure–bending behavior and highlight the distinct deformation profiles of the two designs under identical pressurization (also see Movie 1).

2) Robustness Tests: Robustness under external flow, evaluated in the swim tunnel, shows FSI-optimized actuator mounted in the test section as shown in Supplementary Fig. S3(b) and tested under uniform upstream velocities of 0, 20, and 50 mm/s (see Movie 1). These experimental speeds were selected based on the swim-tunnel operating range and to span quiescent, intermediate, and higher-flow conditions; the 20 mm/s case lies within the simulation verification range of 10–60 mm/s, while 50 mm/s provides a higher-flow robustness case. The corresponding pressure–bending curves are reported in Fig. 4(c). Despite the increased external flow, the actuator preserves nearly identical bending performance across the tested speeds as predicted by simulations. Quantitatively, the maximum deviation in bending angle over the full pressure sweep remains below 2.12% even at 50 mm/s, indicating that the pressure–response characteristics are largely insensitive to moderate upstream flow in this regime. Because the external dimensions of the actuator are fixed, this robustness is not interpreted as a passive drag reduction of an unchanged rigid body. Instead, the internal chamber geometry selected by FSI optimization changes the pressure-driven deformation response. The actuator attains larger bending at a given pressure while the verified coupled drag force on the deformed configuration remains lower than that of the FEM-optimized design at the design condition (Table II). The moderate upstream-flow variation therefore acts as a small perturbation relative to the dominant pressure-driven bending response, which explains why the measured pressure–bending curves remain nearly unchanged across the tested speeds. This result highlights the robustness of the optimized geometry to external disturbances well above those used in optimization.

3) Task-Level Transfer to Underwater Grasping: The FSI-enabled actuator optimization in the preceding sections was aimed at enabling the actuators to envelop objects through less energetic yet enhanced conformal bending. It has been shown from both simulation and experimental results that the FSI-optimized actuators produced larger-angle bending profiles and distributed contact more evenly along the chambered surface. When assembled into a multi-finger configuration, these characteristics translated directly into improved object envelopment during grasping tasks. Both FEM-optimized and FSI-optimized actuators were tested on two representative geometries: a sphere (diameter 10 cm) and a cube (diagonal 10 cm). As shown in Fig. 4(e) and Movie 1, the FSI-optimized gripper wrapped around the objects more rapidly and completely under an input pressure of 20 kPa. In contrast to the FEM-optimized design, the FSI-optimized gripper exhibited greater conformal bending, resulting in increased contact area and enabling stable, reliable lifting of the test objects.

To quantify the gripper-level consequence of the actuator designs, we performed controlled pull force tests with a two-finger gripper and measured the peak retention force before grasp failure. Five objects were used to probe both canonical contact conditions and geometric irregularity: two 3D-printed spheres, two 3D-printed cylinders, and the actuator of Fig. 1(a) as an irregular object. Each was extracted at 10 mm/s under a quasi-steady tensile load at a fixed actuation pressure of 30 kPa, using identical protocols for both assemblies; the

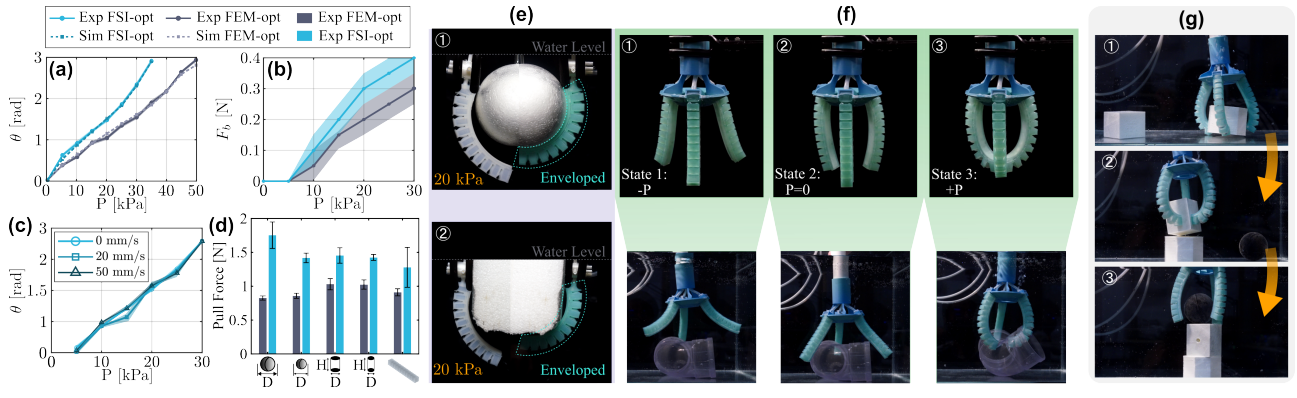

Fig. 4. **Actuator and gripper characterization. (a)** Underwater bending angle versus pressure for FEM-opt and FSI-opt actuators, comparing experimental measurements with external two-way FSI predictions. **(b)** Blocked force F_b versus pressure for both designs. **(c)** Flow robustness of the FSI-opt actuator, showing $\theta(P)$ at $v_s = 0$, 20, and 50 mm/s. **(d)** Peak pull force measured for the FEM-optimized and FSI-optimized actuator assemblies across five test objects at 30 kPa. **(e)** Conformal enveloping of FSI-opt (green) versus FEM-opt (white) actuators around ① a sphere and ② a cube at 20 kPa. **(f)** Pressure-controlled modes: ① gaping, ② standby, and ③ capturing; bottom row demonstrates grasping a large pipe fitting using the gaping-to-capturing transition. **(g)** Underwater stacking sequence of ① an octagonal block, ② a square block, and ③ a sphere.

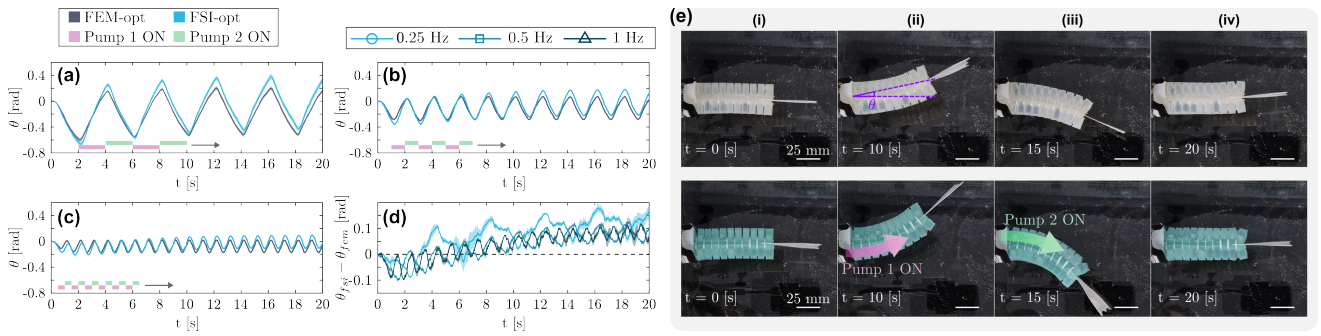

Fig. 5. **Fish-like robot demonstration using paired actuators driven by antagonistic pumping. (a–c)** Measured bending angle $\theta(t)$ under open-loop antagonistic drive at 0.25, 0.5, and 1 Hz; colored bars mark Pump 1 and Pump 2 activation intervals. **(d)** Mean difference $\theta_{\mathrm{FSI}} - \theta_{\mathrm{FEM}}$ over time for the three drive frequencies, with shaded regions denoting ± 1 standard deviation across repeated trials. **(e)** Representative snapshots from a 20 s underwater experiment comparing the FEM-optimized pair (top) and FSI-optimized pair (bottom), illustrating alternating left–right bending.

instrumentation and object dimensions are given in Supplementary Section 1.5 and Fig. S3(c)–(d).

The pull force measurements in Fig. 4(d) show that the FSI-optimized gripper consistently generated higher retention forces than the FEM-optimized gripper across all tested geometries. The improvement ranged from 39% to 112% across the individual objects. Averaged over all five objects, the FSI-optimized design increased the mean pull-out force by 58%. These results indicate that the enhanced conformal bending and contact distribution obtained through FSI-based optimization translate directly into stronger gripper-level retention under underwater loading.

A three-finger gripper was then fabricated using the FSI-optimized actuator geometry to demonstrate practical capability to perform underwater manipulation tasks and the ability to grasp a wide array of irregular objects. The actuators were mounted in a triangular configuration and connected to a single pressure source, with air as the working medium. Grasping was evaluated over a pressure range of 0–35 kPa, consistent with the bearing capacity of the actuators. The gripper could operate in three distinct modes by simply regulating input pressure as illustrated in Fig. 4(f)(①–③): ① gaping (negative pressure, actuators open outward), ② standby (no pressure, actuators hang vertically), and ③ capturing (positive pressure, actuators bend inward to grasp). These modes are particularly useful when handling objects larger than the gripper's effective grasping diameter, since the gaping state allows the fingers to open wider before transitioning to the capturing mode, as demonstrated in the framed sequences of Fig. 4(f) during an underwater grasping test of a large pipe fitting.

Further underwater manipulation experiments, highlighting the gripper's ability to grasp a range of diverse objects, are shown in Fig. 4(g), Supplementary Fig. S4 and Movie 1. Fig. 4(g) demonstrates a stacking task, where the gripper relocated an octagonal block underwater, stacked a square block on top, and finally added a spherical ball. This sequence highlights its ability to execute complex manipulations beyond single-object grasping. Supplementary Fig. S4 and Movie 1 further illustrate grasping and lifting of diverse objects with different sizes, shapes, and stiffnesses. Even objects outside the nominal grasping range were managed by first performing a gaping motion followed by inward closure. Quantitatively, the gripper achieved a maximum payload capacity of 270 g (dry weight) and successfully handled object sizes up to 12 $\times$ 15 cm wide. These demonstrations illustrate the capabilities of the assembled gripper and are not presented as novel tasks. The performance claim is instead supported by the matched tests in Fig. 4(d)–(e), which compare the FSI-optimized assembly with the conventional FEM-optimized baseline summarized in Table III.

4) Fish-Like Robot Demonstration With Paired Actuators: To provide a system-level demonstration beyond single-actuator and gripper tests, we assembled two actuators into a fish-like body and drove them antagonistically using a dual-pump hydraulic circuit. For actuation circuit, hardware design and data collection protocol refer to Supplementary Section 1.6 and Fig. S5(a)–(c). For the paired-actuator tests, each actuation frequency was repeated three times, and the reported time histories are shown as mean $\pm$ standard deviation. Figures 5(a)–(c) report the bending-angle trajectories $\theta(t)$ for the FEM-optimized and FSI-optimized actuator pairs under open-loop antagonistic drive at 0.25, 0.5, and 1 Hz. The colored bars mark the time windows when Pump 1 or Pump 2 is commanded on, so each half-cycle corresponds to pressurization of one side while the opposite side is allowed to deflate. Across all frequencies, the FSI-optimized body attains a larger oscillation amplitude than the FEM-optimized body under the same command schedule, consistent with the actuator-level pressure–bending characterization.

To quantify the experimental dynamic discrepancy between the two designs, Fig. 5(d) plots the tracking difference $\theta_{\mathrm{FSI}} - \theta_{\mathrm{FEM}}$ over time for each drive frequency. Under the same pump commands, the difference remains predominantly positive over successive cycles, indicating larger bending angles for the FSI-optimized body throughout the run. Representative underwater snapshots from a 20 s trial are shown in Fig. 5(e), comparing the FEM-optimized pair (top) and FSI-optimized pair (bottom) at matched time points. Together, the time histories and snapshots show that the FSI-based geometry optimization yields a consistent increase in oscillatory bending in the fish-like assembly under identical open-loop actuation.

To further quantify performance in the fish-like swimming task, we used the DLC-tracked body kinematics as the prescribed actuation input for a two-dimensional FSI analysis,

TABLE III
COMPARISON OF SOFT ACTUATOR STUDIES IN THE LITERATURE.

Ref.	Act. type	FEM	FSI	FSI-opt	FEM-opt	UW	Exp.
[5]	Wavy PneuNet	✓			✓		
[6]	PneuNet		✓				✓
[8]	PneuNet		✓			✓	
[10]	PneuNet	✓			✓		✓
[12]	Bend. Act.	✓			✓		✓
[15]	SPA	✓			✓		
[17]	SPA	✓			✓		
[13]	SPA	✓			✓		✓
[14]	Linear Act.	✓			✓		
[18]	Omni. SPA	✓			✓		
[16]	Bend. Act.	✓			✓		✓
This work	**PneuNet**	✓	✓	✓	✓	✓	✓

with detailed metrics, pressure fields, and velocity-field videos reported in Supplementary Table S7, Supplementary Fig. S6, and Movie 1. This task-level analysis shows that the FSI-optimized body traveled 27.1% farther over the 20 s trial and achieved a 24.8% higher stabilized mean forward velocity than the FEM-optimized body. Importantly, the stabilized net streamwise force changed from positive for the FEM-optimized body, indicating net drag, to negative for the FSI-optimized body, indicating a thrust-producing net force under the same sign convention. Overall, these actuator-, gripper-, and fish-like demonstrations indicate that optimizing geometry with two-way coupled fluid loading produces performance gains that persist in embodied underwater systems.

It should also be noted that improved calibration of the structural FEM model, for example through more extensive material testing or empirical tuning against water-tank data, could reduce absolute prediction error for a given actuator and condition. However, such calibration would not remove the main modeling difference studied here, namely that an FEM-only model does not include the surrounding fluid domain and therefore cannot resolve the geometry-, deformation-, and flow-dependent pressure and shear loads acting on the actuator surface. A geometry-dependent discrepancy model calibrated with coupled or experimental data could, in principle, approximate some of these effects, but it would itself require representative coupled-loading information. By contrast, a uniform post-hoc drag or pressure correction would preserve candidate ordering and therefore cannot reproduce the observed reorderings. This is why the present comparison treats external FSI as part of the optimization data-generation loop rather than as a uniform calibration offset applied after FEM optimization.

Finally, we place the proposed methodology in the context of prior soft actuator design studies. Table III summarizes representative works and highlights whether they employ FEM or two-way FSI, include an explicit FEM- and FSI-optimization (-opt) stage, target underwater (UW) operation, and provide experimental (exp) validation. Prior studies have investigated FSI modeling of soft actuators [6]–[8] and separate optimization strategies such as FEM-based parameter optimization [10], topology optimization [12]–[14] or surrogate-based design [15], [17], [18]. Our work bridges this divide by using FSI-derived responses as surrogate training data, comparing the resulting geometry directly against a FEM-derived counterpart under matched conditions, and validating both designs experimentally through actuator characterization, robustness testing, and system-level underwater demonstrations.

VI. CONCLUSION

This paper presented an external-fluid–structure-interaction-in-the-loop pipeline for geometric optimization of underwater soft fluidic actuators. A baseline comparison demonstrated that conventional structural finite element analysis and two-way coupled simulation can yield significantly different pressure–deformation predictions for the same geometry, motivating coupled modeling for underwater design. Using identical design-of-experiments sampling, surrogate fitting, and objective formulation, we compared geometry selected by FEM-only data against geometry selected by two-way-coupled data, verifying both with high-fidelity FSI. The FSI-derived optimum reduced strain energy and peak stress, increased bending angle and tip displacement, and produced a lower coupled drag force relative to the FEM-derived optimum under matched conditions; water-tank experiments corroborated the pressure–bending and blocked-force trends, and robustness tests showed negligible sensitivity to external flow up to $5\times$ the design speed. Gripper pull-force tests and paired-actuator fish experiments further confirmed that the performance advantage of the FSI-optimized geometry extends beyond individual actuator behavior to practical underwater manipulation and locomotion tasks. Future work will extend the framework to higher-dimensional design spaces and incorporate task-level objectives for manipulation and locomotion under unsteady flow.

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VII. Biography Section

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Guangming Xie (*Senior Member, IEEE*) is a Professor of Dynamics and Control with the School of Advanced Manufacturing and Robotics, Peking University. He received his Ph.D. in Control Theory and Control Engineering from Tsinghua University in 2001.

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Supplementary Material for

An Optimization Framework for Soft Fluidic Actuators: Fully Coupled FSI Simulation with Surrogate Modeling

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1. Supplementary Text

1.1. Hyperelastic Material Model Derivation and Mechanical Property Test

Hyperelastic framework SFAs undergo large deformations, so the silicone elastomer is modeled as hyperelastic materials. The deformation state is described by the principal stretch ratios $\lambda_i = l_i/L_i$ ($i \in \{1, 2, 3\}$), defined as the ratio of deformed to undeformed length along each principal axis [1]. The principal invariants of Cauchy–Green deformation tensor are expressed as

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \tag{1}$$
$$I_2 = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_1^2\lambda_3^2, \tag{2}$$
$$I_3 = \lambda_1^2\lambda_2^2\lambda_3^2. \tag{3}$$

For isotropic, incompressible hyperelastic materials, $I_3 = 1$, the principal Cauchy stress follows from the strain energy density W via virtual work considerations [2, 3]:

$$\sigma = 2\left(\lambda_1^2\frac{\partial W}{\partial I_1} - \lambda_1^{-2}\frac{\partial W}{\partial I_2}\right) + p, \tag{4}$$

where p is the hydrostatic pressure (a Lagrange multiplier determined from equilibrium and boundary conditions) [4, 5]. The general N-term Yeoh strain energy density is

$$W = \sum_{i=1}^{N} C_{i0}\,(\bar{I}_1 - 3)^i + \sum_{k=1}^{N}\frac{1}{D_k}(J-1)^{2k}, \tag{5}$$

where J is the volume ratio between deformed and undeformed states (equal to 1 for incompressible materials), and $\bar{I}_1 = J^{-2/3}I_1$. The constants C_{i0} and D_k are material parameters and require experimentation to be determined [6]. In the main manuscript a third-order Yeoh model ($N = 3$, $D_1 = D_2 = D_3 = 0$) is used, reducing to $W = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2 + C_{30}(\bar{I}_1 - 3)^3$.

Mechanical property tests To characterize the mechanical properties, uniaxial tension tests were conducted on ASTM-standard dogbone specimens. The tests were executed using a uniaxial testing machine (Mark-10 F105) equipped with a 100 N load cell (Fig. S1(a)). The resulting stress-strain response up to failure (at a strain of 6.2) is depicted in Fig. S1(b). The observed

material behavior was found to be accurately captured by the Yeoh model. The fitted model coefficients are $C_{10} = 0.06066$, $C_{20} = 0.00224$, $C_{30} = -1.8532 \times 10^{-5}$.

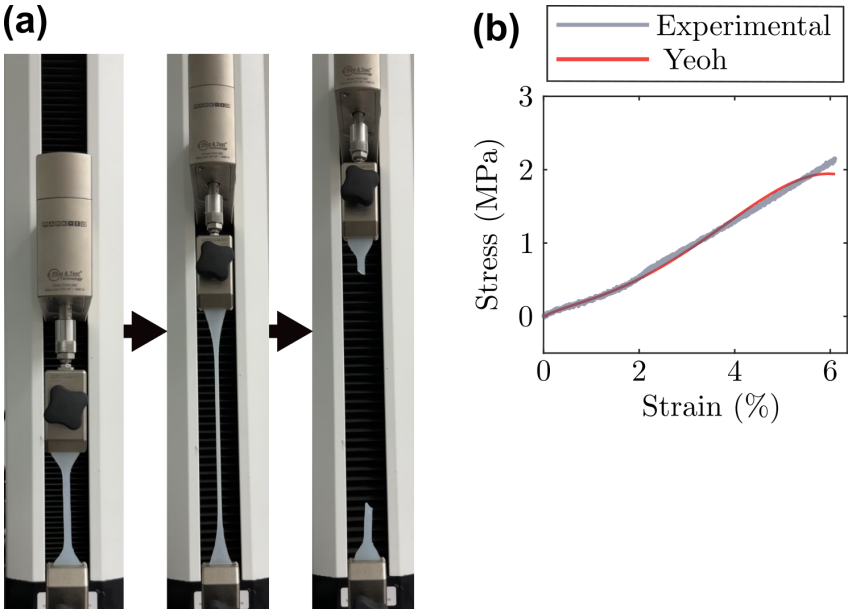

Fig. S1: Material characterization test. **(a)** Uniaxial tensile testing of the DragonSkin-20 specimen using a Mark-10 F105 test stand. **(b)** Measured stress–strain curve with fitted Yeoh hyperelastic model.

1.2. Mesh and Time-Step Sensitivity Analysis

To verify that the numerical settings used for the DOE were sufficiently resolved, mesh and time-step sensitivity studies were conducted at the design pressure, with results summarized in Table S1. Mesh sensitivity was evaluated for both the baseline and FSI-optimized geometries using coarse (48823 solid elements, 267737 fluid elements), medium (140667 solid elements, 742900 fluid elements), and fine (832376 solid elements, 3702285 fluid elements) discretizations. Time-step sensitivity was evaluated using the FSI-optimized geometry with the medium mesh. The monitored quantities were the same responses used in the optimization: bending angle, tip displacement, maximum von Mises stress, and strain energy.

Table S1: Mesh and time-step sensitivity of the FSI model at $10kPa$.

Case	Setting	θ (rad)	Y_{tip} (mm)	$\sigma_{\max}$ (MPa)	U (mJ)
Baseline	Coarse	0.393	-52.988	0.059	10.676
Baseline	Medium	0.414	-57.922	0.065	11.263
Baseline	Fine	0.416	-58.553	0.070	11.358
FSI-opt	Coarse	0.4339	-67.367	0.097	15.856
FSI-opt	Medium	0.4244	-60.942	0.119	12.787
FSI-opt	Fine	0.4250	-61.145	0.116	12.892
FSI-opt	$\Delta t = 0.005$ s	0.426	-61.260	0.116	12.897
FSI-opt	$\Delta t = 0.010$ s	0.424	-60.942	0.119	12.787
FSI-opt	$\Delta t = 0.020$ s	0.425	-61.127	0.116	12.891
FSI-opt	$\Delta t = 0.040$ s	0.425	-61.117	0.116	12.890
FSI-opt	$\Delta t = 0.080$ s	0.422	-60.271	0.116	12.826

Refining the baseline mesh from medium to fine changed the bending angle by 0.5%, tip displacement by 1.1%, and strain energy by 0.8%. For the FSI-optimized geometry, the corresponding changes were 0.1%, 0.3%, and 0.8%, respectively. The maximum-stress variation from medium to fine mesh was 7.1% for the baseline and 2.6% for the FSI-optimized geometry. Time-step sensitivity using the FSI-optimized geometry showed that $\Delta t = 0.01$ s produced nearly unchanged deformation and energy responses relative to smaller and larger stable steps. Therefore, the medium mesh and $\Delta t = 0.01$ s were retained for the DOE and optimization studies.

Although Table S1 shows little accuracy gain from $\Delta t = 0.01$ s over coarser steps up to 0.080 s for the tested geometry, this check alone does not justify using a coarser step for the full workflow. Several of the more compliant DOE geometries failed to converge at time steps larger than $\Delta t = 0.01$ s, as the larger per-step displacement exceeded what the dynamic-mesh remeshing could resolve within the allowed coupling iterations. Thus, $\Delta t = 0.01$ s was retained not only for accuracy, but because it was the largest step that converged reliably across the entire 90-point DOE set.

The Mechanical–Fluent exchange was implemented as a partitioned implicit, staggered System Coupling procedure: within each time step, the solvers were advanced sequentially and iterated until interface-data convergence or the prescribed maximum number of coupling iterations was reached. Default System Coupling relaxation settings were used; no user-specified under-relaxation factor was imposed. The system-coupling convergence history was also monitored using the root-mean-square change of the transferred interface data. The default 10^{-3} system coupling convergence criteria were used. The RMS changes of the structural and fluid data-transfer quantities remained bounded and generally decreased to small values over the coupling iterations, indicating stable two-way data exchange between Fluent and Mechanical.

1.3. Objective-weight Sensitivity Analysis

To examine whether the DOE-level ranking was an artifact of the chosen scalarization weights, we re-evaluated the existing 10 kPa DOE points using six representative weighting schemes, including balanced, stress-conservative, energy-conservative, angle-dominant, and displacement-dominant cases (Table S2; Fig. S2(a)). This check was not intended to locate the final optimum; rather, it tested whether the preferred DOE candidate changed when the relative importance of the objective terms was varied before response-surface construction and continuous surrogate-based optimization. Because surrogate-family and optimizer choices were examined separately, this analysis was kept as a focused scalarization check rather than a full re-optimization for every weight set. In all six manual weighting cases, the same evaluated DOE candidate, DP 54, was selected. We further sampled 10,000 random weight vectors from a uniform Dirichlet distribution over the four objective terms (Fig. S2(b)–(c)). DP 54 remained the most frequently selected candidate, accounting for 60.6% of all random weighting cases, followed by DP 10 and DP 7 with 14.5% and 12.5%, respectively. These results indicate that, within the sampled DOE set, the DOE-level ranking is robust to the objective-weight choice, while the final optimized geometry is obtained separately from the response-surface-based continuous optimization.

Table S2: Manual objective-weight sensitivity summary at the design pressure of 10 kPa.

Weighting scheme	w_d	w_θ	w_σ	w_U	Selected DP	Minimum F
Original	0.40	0.40	0.10	0.10	54	0.137
Balanced	0.25	0.25	0.25	0.25	54	0.343
Stress-conservative	0.30	0.30	0.30	0.10	54	0.212
Energy-conservative	0.30	0.30	0.10	0.30	54	0.337
Angle-dominant	0.20	0.50	0.15	0.15	54	0.206
Displacement-dominant	0.50	0.20	0.15	0.15	54	0.206

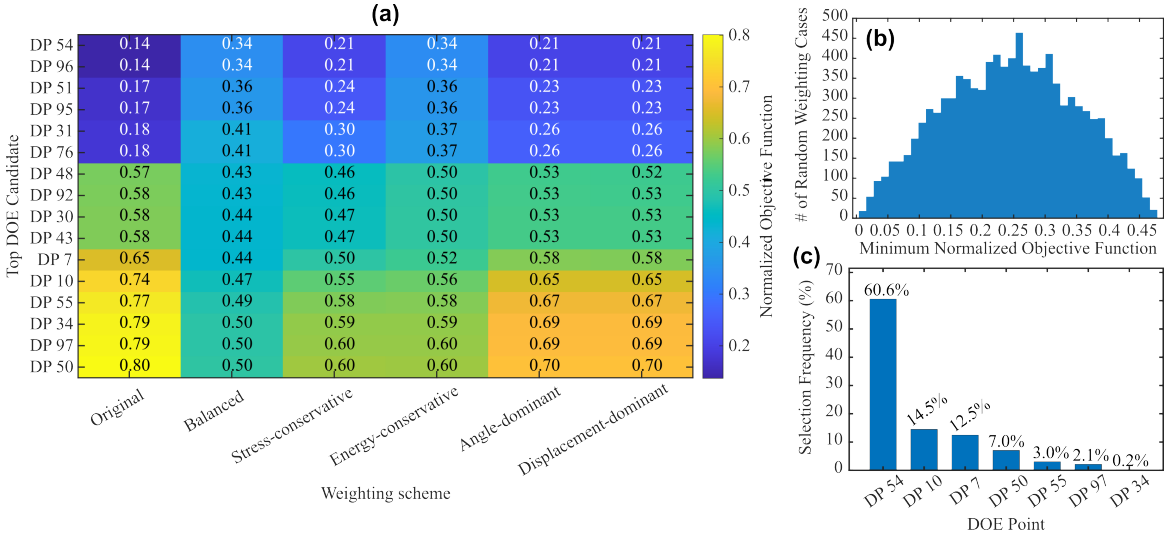

Fig. S2: Objective-weight sensitivity analysis at the design pressure of 10 kPa. (**a**) The normalized scalar objective values for the top DOE candidates under six representative weighting schemes. The color scale represents the normalized objective value, where lower values indicate better performance. (**b**) The distribution of the minimum normalized objective values obtained from 10,000 randomly sampled weighting combinations, with the weights constrained by $w_d + w_\theta + w_\sigma + w_U = 1$. (**c**) The DOE-point selection frequency from the Monte Carlo weighting analysis.

Table S3: Key implementation details of the paired FEM/FSI surrogate-optimization workflow.

Item	Detail
Design variables	t_{w1}, t_{w2}, $w_c \in [0.5, 3.0]$ mm
DOE	90 identical geometries for FEM-only and FSI datasets
Condition	$P_d = 10$ kPa, $v_s = 0.01$ m/s
Mesh/time step	1 mm solid, 2 mm fluid, $\Delta t = 0.01$ s
Responses	Y_{tip}, θ, $\sigma_{\max}$, U, drag
Objective	Maximize Y_{tip}, θ; minimize $\sigma_{\max}$, U
Weights	$w_d = 0.4$, $w_\theta = 0.4$, $w_\sigma = 0.1$, $w_U = 0.1$
Surrogate	Non-parametric regression selected by validation nRMSE
Comparison	FEM-opt and FSI-opt re-evaluated by high-fidelity FSI

Table S4: Pressure and flow conditions used in simulation, optimization, and experiments.

Study component	Pressure condition	Flow condition
DOE and surrogate training	$P_d = 10$ kPa	$v_s = 0.01$ m/s
Objective-weight sensitivity	$P_d = 10$ kPa	$v_s = 0.01$ m/s
Mesh and time-step verification	$P_d = 10$ kPa	$v_s = 0.01$ m/s
FEM-opt/FSI-opt verification	$P_d = 10$ kPa	$v_s = 0.01$, 0.02, 0.04, 0.06 m/s
Bending-angle experiment	0–50 kPa sweep	Quiescent water
Blocked-force experiment	0–50 kPa sweep	Quiescent water
Flow-robustness experiment	0–50 kPa sweep	$v_s = 0$, 20, 50 mm/s
Enveloping experiment	20 kPa (fixed)	Quiescent water
Pull-force experiment	30 kPa (fixed)	Quiescent water
Fish-like body demonstration	Periodic pressure input	Quiescent water

1.4. Cross-Evaluation of DOE Candidates under FEM-only and Coupled Data

Table II of the main text establishes that the FEM-only and coupled workflows converge on different geometries. This section identifies the mechanism at the sampled-data level. For a fixed set of candidates and a fixed scalar objective, modeling fidelity changes the selected design only when it changes the objective ordering. Table S5 therefore scores the same sampled geometries under the FEM-only and two-way FSI datasets using the scalar objective F of Eq. (4) of the main text: the weighted sum of the min–max normalized tip displacement, bending angle, maximum von Mises stress, and strain energy. Within each material comparison, the geometries, normalization ranges, and objective weights are identical, so the only changed input is whether the surrounding fluid is resolved. Sampled candidates are used because their values come directly from paired simulations and contain no surrogate-interpolation contribution.

Across the paired DragonSkin 20 candidates, approximately 50% of all candidate pairs are ordered differently by the two datasets. For reference, identical rankings would produce no discordant pairs; approximately half being reordered indicates that the FEM-only ranking provides essentially no monotonic guide to the coupled ranking. The displayed pair illustrates this reordering: FEM-only data prefer D1, whereas the coupled data prefer D2. The Elastosil M4601 material-sensitivity replication gives the same qualitative result: the two datasets again prefer different sampled geometries and reorder a substantial fraction of candidate pairs. The effect is therefore confined neither to the displayed pair nor to one elastomer.

The reordering is geometry dependent because the fluid loading acts on the pressurized shape, which varies with wall thickness and channel width. Omitting the fluid therefore does not produce a uniform offset or an order-preserving rescaling of the sampled response surface. It changes the relative merit of candidate geometries and can consequently displace the minimizer. This result does not preclude a geometry-dependent discrepancy model calibrated with coupled or experimental data; rather, it shows why a uniform post-hoc correction to a structural response cannot reproduce the observed ordering.

Table S5: Cross-evaluation of sampled geometries under FEM-only and two-way FSI data at 10 kPa and $v_s = 0.01$ m/s. Dimensions are (t_{w1}, t_{w2}, w_c) in mm: D1=$(2.313, 0.687, 0.734)$, D2=$(1.500, 0.500, 1.750)$, E1=$(0.687, 0.687, 2.766)$, and E2=$(2.313, 0.687, 0.734)$. Within each material, both datasets use identical geometries, normalization ranges, and objective weights. Lower F is better.

Material	Evaluation	First candidate	Second candidate	Selects
DragonSkin 20	FEM-only F	**0.164** (D1)	0.264 (D2)	D1
	Two-way FSI F	0.558 (D1)	**0.221** (D2)	D2
Elastosil M4601	FEM-only F	0.727 (E1)	**0.200** (E2)	E2
	Two-way FSI F	**0.180** (E1)	0.705 (E2)	E1

Table S6 complements this DOE-level comparison with the final Elastosil M4601 geometries obtained from the two surrogate-optimization workflows and their matched two-way FSI verification. The FSI-selected geometry is distinct from the FEM-selected geometry and improves all five reported verification responses.

Table S6: Material-sensitivity check using Elastosil M4601 at 10 kPa and $v_s = 0.01$ m/s.

Design	t_{w1} mm	t_{w2} mm	w_c mm	U mJ	$\sigma_{\max}$ MPa	Y_{tip} mm	Drag N	θ rad
Baseline	1.0	1.0	2.0	11.269	0.0730	-42.448	8.8×10^{-6}	0.408
FEM-opt	2.0	2.1	1.8	8.013	0.0678	-46.336	8.2×10^{-6}	0.453
FSI-opt	2.0	1.5	2.5	6.179	0.0417	-57.998	3.69×10^{-6}	0.604
FSI-opt vs FEM-opt	–	–	–	-22.9%	-38.5%	+25.2%	-55.0%	+33.3%
FSI-opt vs Baseline	–	–	–	-45.2%	-42.9%	+36.6%	-58.1%	+48.0%

1.5. Experimental Setup for Actuator Evaluation

A test platform was made to validate the simulation results and quantify actuator performance. Two environments were considered: stationary water tests in a ($0.2 \times 0.12 \times 0.12$ m) glass tank, and dynamic flow tests in a recirculating Loligo swim tunnel with test section $0.25 \times 0.25 \times 0.87$ m (Fig. S3(a)–(b)). The Loligo setup consists of the SFA, a mounting fixture, a pressurization circuit with a manual regulating valve, a pressure sensor (RSCM17100KP201-V01, Shenzhen Dahua Electronics Co., China), a high-speed camera (Sony RX100m5) for motion capture, and a digital force meter (HP-50, Yueqing Hampi Instrument Company, China; resolution: 0.01 N, error: $\pm 0.5\%$) for blocked force measurements. Owing to the segmented chamber structure, which reduces stiffness in the bending plane, the actuator was mounted horizontally with its bending plane parallel to the ground to minimize gravitational effects. To keep the stationary- and moving-water tests comparable, the actuator was tested at the same submergence depth in both setups; this depth was selected based on the Loligo swim-tunnel test section, which constrained the dynamic-flow measurements.

Internal pressure was manually controlled between 0–50 kPa in 5 kPa increments. For dynamic tests in the swim tunnel, flow velocities of 0, 20, and 50 mm/s were imposed. Flow velocities were measured at location of actuator in the test section using a turbine-type flow probe (LS300A portable velocity meter 0.01-4.0 m/s). In all cases, actuator motion was recorded perpendicular to the bending plane, and Fiji (NIH, USA) was used to extract bending angles from tip trajectories offline. Each condition was repeated three times to obtain mean and standard deviation values.

During bending tests in quiescent water, the actuator inlet was rigidly clamped and the distal end was left unconstrained. For each pressure level, the pressure was increased to the target value, held until the deformation stabilized, and the final configuration was recorded to extract the bending angle. These repetitions help characterize experimental variability arising from silicone batch differences, curing conditions, wall-thickness deviations, bonding quality, leakage, and trapped-air effects, which are not represented by simulations using nominal hyperelastic parameters.

The pull-force measurements used five target objects selected to probe both canonical contact conditions and geometric irregularity: two 3D-printed spheres (40 and 50 mm in diameter), two 3D-printed cylinders (56 mm in height, with diameters of 30 and 40 mm), and the actuator geometry itself as an irregular test object. For each trial, the grasped object was tethered by an inextensible fishing wire to a DECCA DK-F2S (0.2 kg) digital push–pull force gauge (rated capacity: 2 N/0.2 kg; accuracy: $\pm 1\%$; resolution: 0.001 N) mounted on a FUYU FSL30-series motorized linear module with a 150 mm travel distance (Fig. S3(c)–(d)). The force gauge was zeroed before each trial, and the linear stage then translated at 10 mm/s, imposing a quasi-steady tensile load until the object was extracted from the grasp. Identical protocols were used for both actuator assemblies, with the actuation pressure fixed at 30 kPa throughout the measurements.

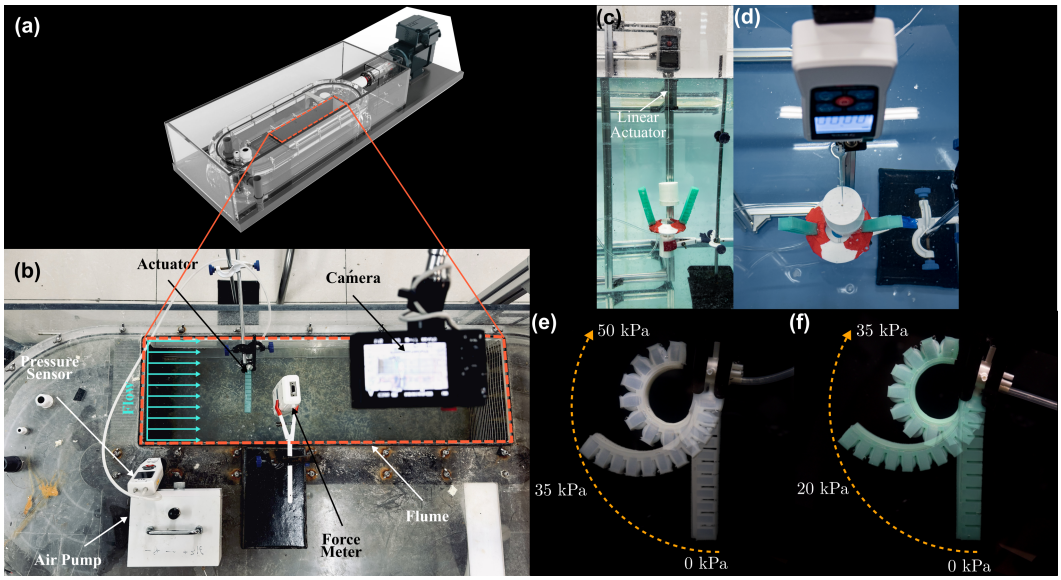

Fig. S3: **(a–b)** Loligo swim-tunnel setup and instrumented test section showing the actuator, pressure supply, pressure sensor, force meter, camera, and upstream flow direction. **(c)** Experimental setup showing the linear actuator used to impose a controlled tensile displacement on the grasped object in water. **(d)**. Top-view photograph of a representative pull-out trial with the object tethered to the load cell and grasped by the two-finger soft gripper. **(e–f)** Representative deformation snapshots under increasing pressure for the FEM-opt actuator (white) and FSI-opt actuator (green).

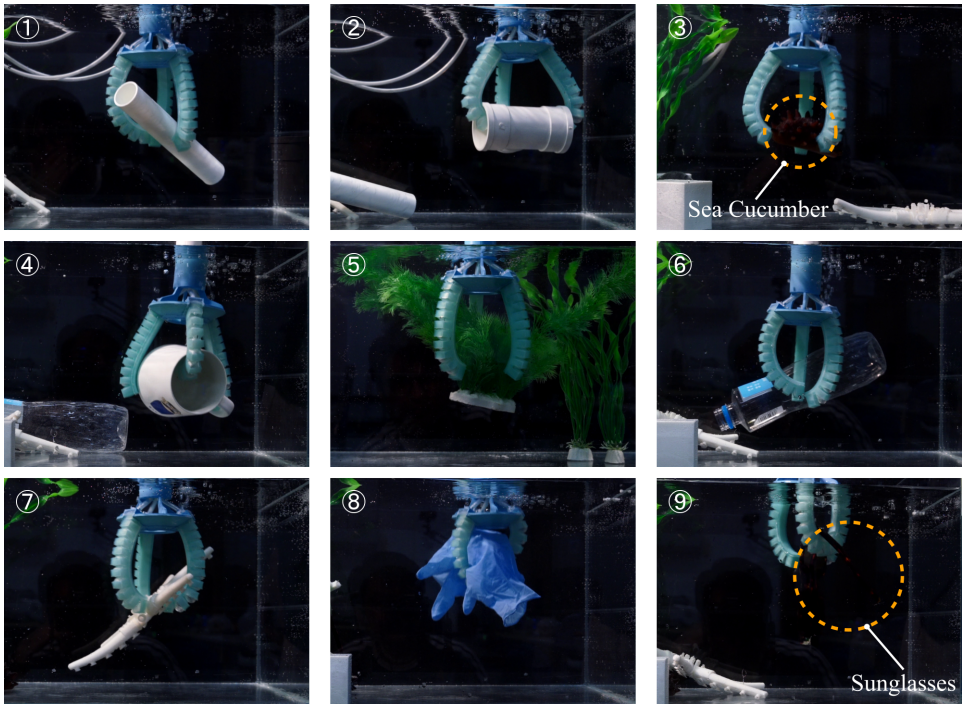

Fig. S4: Representative grasps of varied objects (rigid and compliant), including ① PVC pipe, ② T-fitting, ③ sea cucumber, ④ ceramic cup, ⑤ underwater plant, ⑥ plastic bottle, ⑦ starfish-shaped object, ⑧ latex gloves, and ⑨ sunglasses.

1.6. Fish-Like robot setup and data collection protocol

The control and actuation architecture is summarized in Fig. S5(a). The actuation system used two geared pumps (GGMN 1200, Korea) with water as the working fluid, driven through a dual H-bridge motor driver (L298N). The pumps were driven with an open-loop PWM command of 200 (on an 8-bit 0–255 scale) and alternated to pressurize one side while the opposing side was allowed to deflate, producing left–right bending. The experimental setup and underwater test tank are shown in Fig. S5(b)–(c), where the body was clamped at the head and the tail was left free to oscillate. We extracted bending angle using DeepLabCut (DLC) [7] from 60 fps video recordings. The DLC training set comprised two videos, one per fish, recorded at 0.25 Hz actuation; 20 frames were selected by k-means and manually labeled with head or clamped-base and tail keypoints. We trained a ResNet-50 model for 50,000 iterations using default DLC settings. During inference, only poses with likelihood > 0.9 were retained and visually verified by overlay on the original frames (Movie 1).

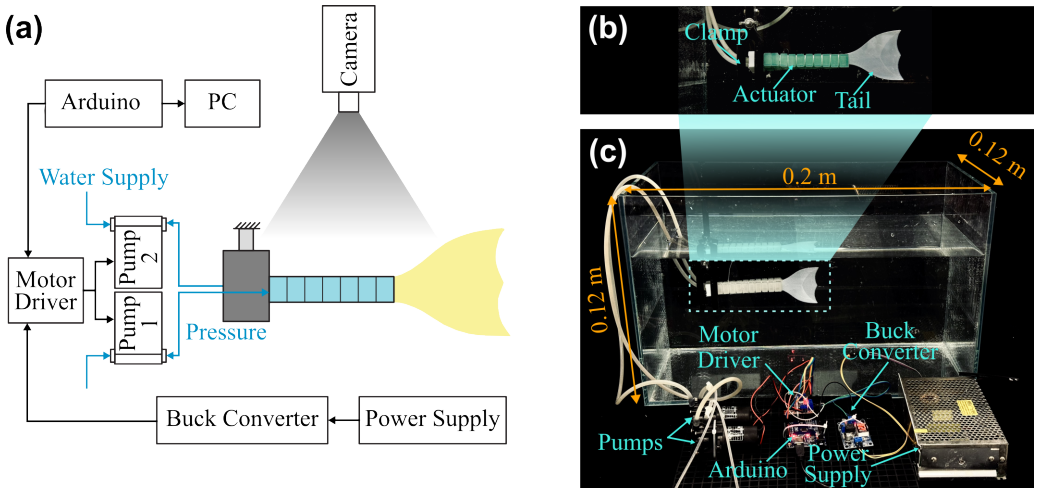

Fig. S5: Fish-like robot demonstration using paired actuators driven by antagonistic pumping. **(a)** Control and actuation schematic. **(b,c)** Water-tank setup with the paired-actuator body clamped at the head; key components and dimensions shown.

1.7. DLC-Driven Quantification of Swimming-Task Performance

To further quantify performance in the fish-like swimming task, we used the experimentally measured DLC kinematics as prescribed motion input for two-dimensional FSI simulations. Twelve DLC points along each body were used to drive the deforming midline, and the surrounding water domain was solved to compare displacement, swimming speed, pressure field, velocity field, and streamwise hydrodynamic force. This analysis is intended as a task-level, kinematics-driven comparison of the two experimentally observed swimming motions.

The experimental recordings were 20 s long. We therefore report displacement at both 10 s and 20 s, but compute the mean forward velocity and net streamwise force over the 10–20 s interval. This avoids using the initial start-up portion of the motion, during which the body, flow field, and actuation response are still approaching a repeatable oscillatory state. Using the later half of the trial provides a more conservative and comparable estimate of stabilized swimming-task performance for the two designs.

The quantitative results in Table S7 show that the FSI-opt fish-like body travels farther than the FEM-opt body under the same DLC-driven actuation, producing 34.6% larger displacement at 10 s and 27.1% larger displacement at 20 s. Over the stabilized 10–20 s interval, its mean forward velocity is 24.8% higher. The sign of the net streamwise force is particularly important:

Table S7: DLC-driven swimming-task performance comparison of FEM-opt and FSI-opt fish-like bodies. Mean forward velocity and net streamwise force are computed over the stabilized 10–20 s interval. Positive net streamwise force denotes drag, while a negative value denotes a thrust-producing net force under the simulation sign convention.

Metric	FEM-opt fish	FSI-opt fish	FSI-opt change
Distance at 10 s (mm)	19.626	26.410	+34.6%
Distance at 20 s (mm)	81.966	104.220	+27.1%
Mean forward velocity (mm s^{-1})	6.234	7.781	+24.8%
Net streamwise force (N)	0.001843	-0.00237545	Thrust producing

the FEM-opt body retains a positive net force, indicating net drag, whereas the FSI-opt body produces a negative net force under the same sign convention, indicating that the integrated hydrodynamic force is thrust producing. Thus, the swimming-task comparison is consistent with the main actuator-level result: the FSI-optimized geometry produces larger body motion and a more favorable hydrodynamic force balance under matched kinematic extraction and analysis conditions.

The pressure fields in Fig. S6 provide a qualitative explanation for this force balance. During the larger-amplitude portions of the stroke, the fish-like body generates alternating high- and low-pressure regions around the deforming tail and actuator body. These pressure differences are strongest when the body curvature and lateral sweep are largest, and they relax during the more nearly straight configurations. This phase-dependent pressure asymmetry is consistent with the sign change in the integrated streamwise force reported in Table S7: the FSI-opt motion produces a net hydrodynamic contribution in the thrust direction over the stabilized interval, whereas the FEM-opt motion remains drag dominated. The corresponding velocity-field evolution and robot motion are shown in Movie 1.

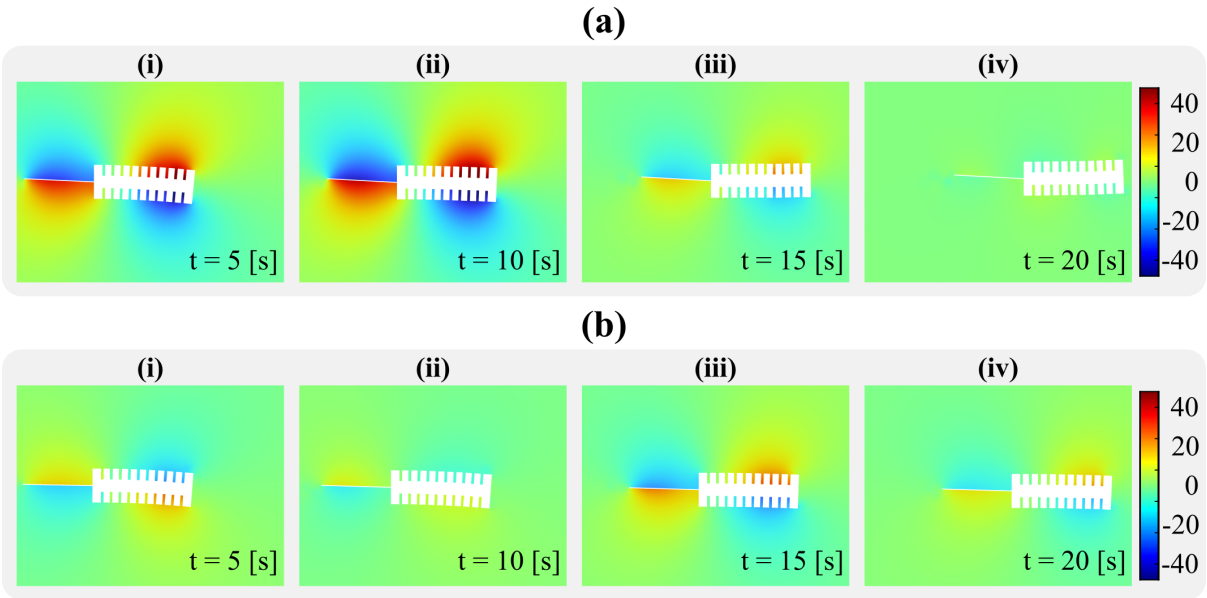

Fig. S6: Pressure-field snapshots from the DLC-driven swimming-task comparison of fish-like bodies at four representative instants. Row (a) shows the FSI-opt fish-like body and row (b) shows the FEM-opt fish-like body at corresponding instants.

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